

数值分析第三次作业

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1. 求证: $f[x_0, x_1, \dots, x_k]$ 是 $f_0, f_1, \dots, f_k$ 的线性组合, 进一步有

$$f[x_0, x_1, \dots, x_k] = \sum_{j=0}^k \frac{f(x_j)}{\omega'_{k+1}(x_j)}, \quad \omega_{k+1}(x) := \prod_{j=0}^k (x - x_j).$$

解答.

只需证:

$$f[x_0, x_1, \dots, x_k] = \sum_{j=0}^k \frac{f(x_j)}{\omega'_{k+1}(x_j)}.$$

对 $k \in \mathbb{Z}_{\geq 0}$ 归纳, 当 $k = 0$ 时平凡, 假设命题对 $k \in \mathbb{Z}_{\geq 0}$ 成立, 对 $k + 1$: 由归纳假设,

$$\begin{aligned} f[x_0, x_1, \dots, x_k] &= \sum_{j=0}^k \frac{f(x_j)}{\alpha'_{k+1}(x_j)}, \quad \alpha_{k+1}(x) := \prod_{j=0}^k (x - x_j), \\ f[x_1, x_2, \dots, x_{k+1}] &= \sum_{j=1}^{k+1} \frac{f(x_j)}{\beta'_{k+1}(x_j)}, \quad \beta_{k+1}(x) := \prod_{j=1}^{k+1} (x - x_j), \end{aligned}$$

故

$$\begin{aligned} & f[x_0, x_1, \dots, x_k, x_{k+1}] \\ &= \frac{1}{x_{k+1} - x_0} (f[x_1, x_2, \dots, x_{k+1}] - f[x_0, x_1, \dots, x_k]) \\ &= \frac{1}{x_{k+1} - x_0} \left(\sum_{j=1}^{k+1} \frac{f(x_j)}{\beta'_{k+1}(x_j)} - \sum_{j=0}^k \frac{f(x_j)}{\alpha'_{k+1}(x_j)} \right) \\ &= \frac{1}{x_{k+1} - x_0} \left(\frac{f(x_0)}{-\alpha'_{k+1}(x_0)} + \sum_{j=1}^k f(x_j) \left(\frac{1}{\beta'_{k+1}(x_j)} - \frac{1}{\alpha'_{k+1}(x_j)} \right) + \frac{f(x_{k+1})}{\beta'_{k+1}(x_{k+1})} \right), \end{aligned}$$

从而只需证:

$$-\frac{1}{(x_{k+1} - x_0)\alpha'_{k+1}(x_0)} = \frac{1}{\omega'_{k+2}(x_0)}, \quad \frac{1}{(x_{k+1} - x_0)\beta'_{k+1}(x_{k+1})} = \frac{1}{\omega'_{k+2}(x_{k+1})},$$

$$\frac{1}{x_{k+1} - x_0} \left(\frac{1}{\beta'_{k+1}(x_j)} - \frac{1}{\alpha'_{k+1}(x_j)} \right) = \frac{1}{\omega'_{k+2}(x_j)}, \quad \forall 1 \leq j \leq k.$$

事实上,

$$\begin{aligned} -\frac{1}{(x_{k+1} - x_0)\alpha'_{k+1}(x_0)} &= \frac{1}{(x_0 - x_{k+1}) \prod_{j=1}^k (x_0 - x_j)} = \frac{1}{\prod_{j=1}^{k+1} (x_0 - x_j)} = \frac{1}{\omega'_{k+2}(x_0)}, \\ \frac{1}{(x_{k+1} - x_0)\beta'_{k+1}(x_{k+1})} &= \frac{1}{(x_{k+1} - x_0) \prod_{j=1}^k (x_{k+1} - x_j)} = \frac{1}{\prod_{j=0}^k (x_{k+1} - x_j)} = \frac{1}{\omega'_{k+2}(x_{k+1})}, \\ \frac{1}{x_{k+1} - x_0} \left(\frac{1}{\beta'_{k+1}(x_j)} - \frac{1}{\alpha'_{k+1}(x_j)} \right) &= \frac{1}{x_{k+1} - x_0} \left(\frac{1}{\prod_{i=1, i \neq j}^{k+1} (x_j - x_i)} - \frac{1}{\prod_{i=0, i \neq j}^k (x_j - x_i)} \right) \\ &= \frac{1}{\prod_{i=1, i \neq j}^k (x_j - x_i)} \cdot \frac{1}{x_{k+1} - x_0} \left(\frac{1}{x_j - x_{k+1}} - \frac{1}{x_j - x_0} \right) \\ &= \frac{1}{\prod_{i=1, i \neq j}^k (x_j - x_i)} \cdot \frac{1}{(x_j - x_{k+1})(x_j - x_0)} \\ &= \frac{1}{\prod_{i=0, i \neq j}^{k+1} (x_j - x_i)} \\ &= \frac{1}{\omega'_{k+2}(x_j)}, \quad \forall 1 \leq j \leq k. \end{aligned}$$

因此命题对 $k+1$ 也成立, 由数学归纳法知命题得证. □

2. 定义向前差分算子

$$\nabla^+ f(x) = f(x+h) - f(x), \quad (\nabla^+)^k = \nabla^+((\nabla^+)^{k-1}), \quad \forall k \geq 2,$$

求证:

$$(\nabla^+)^k f(x_0) = k! h^k f[x_0, x_1, \dots, x_k],$$

其中 $x_j = x + jh, 0 \leq j \leq k$. 对向后差分算子 $\nabla^- f(x) = f(x) - f(x-h)$ 有类似结论.

解答.

对 $k \in \mathbb{Z}_{\geq 0}$ 归纳, 当 $k=0$ 时平凡, 假设命题对 $k \in \mathbb{Z}_{\geq 0}$ 成立, 对 $k+1$: 由归纳假设

$$(\nabla^+)^k f(x_0) = k! h^k f[x_0, x_1, \dots, x_k], \quad (\nabla^+)^k f(x_1) = k! h^k f[x_1, x_2, \dots, x_{k+1}],$$

故

$$\begin{aligned}
 & (\nabla^+)^{k+1} f(x_0) \\
 &= \nabla^+((\nabla^+)^k f(x_0)) \\
 &= (\nabla^+)^k f(x_1) - (\nabla^+)^k f(x_0) \\
 &= k!h^k f[x_1, x_2, \dots, x_{k+1}] - k!h^k f[x_0, x_1, \dots, x_k] \\
 &= k!h^k \cdot \frac{f[x_1, x_2, \dots, x_{k+1}] - f[x_0, x_1, \dots, x_k]}{x_{k+1} - x_0} \cdot (k+1)h \\
 &= (k+1)!h^{k+1} f[x_0, x_1, \dots, x_k, x_{k+1}],
 \end{aligned}$$

因此命题对 $k+1$ 也成立, 由数学归纳法知命题得证. □

3. 对 2 点 3 次 Hermite 插值用基函数表示:

$$H(x) = y_0^{(0)} h_0^0(x) + y_0^{(1)} h_0^1(x) + y_1^{(0)} h_1^0(x) + y_1^{(1)} h_1^1(x),$$

其中

$$\begin{cases} h_i^0(x_j) = \delta_{ij}, & (h_i^0)'(x_j) = 0, \\ h_i^1(x_j) = 0, & (h_i^1)'(x_j) = \delta_{ij}, \end{cases} \quad i, j = 0, 1.$$

写出 $h_i^j(x), i, j = 0, 1$.

解答.

对于 $h_0^0(x)$, 由于

$$h_0^0(x_0) = 1, \quad h_0^0(x_1) = (h_0^0)'(x_0) = (h_0^0)'(x_1) = 0,$$

故 x_1 为 $h_0^0(x)$ 的 2 阶零点, 设 $h_0^0(x) = (ax+b)(x-x_1)^2$, 从而

$$(h_0^0)'(x) = a(x-x_1)^2 + 2(ax+b)(x-x_1),$$

代入 $h_0^0(x_0) = 1, (h_0^0)'(x_0) = 0$ 解得

$$a = -\frac{2}{(x_0 - x_1)^3}, \quad b = \frac{3x_0 - x_1}{(x_0 - x_1)^3},$$

因此

$$h_0^0(x) = \left(1 - 2\frac{x-x_0}{x_0-x_1}\right) \left(\frac{x-x_1}{x_0-x_1}\right)^2,$$

同理可得

$$h_1^0(x) = \left(1 - 2\frac{x - x_1}{x_1 - x_0}\right) \left(\frac{x - x_0}{x_1 - x_0}\right)^2.$$

对于 $h_0^1(x)$, 由于

$$h_0^1(x_0) = h_0^1(x_1) = (h_0^1)'(x_1) = 0, \quad (h_0^1)'(x_0) = 1,$$

故设 $h_0^1(x) = c(x - x_0)(x - x_1)^2$, 从而

$$(h_0^1)'(x) = c(x - x_1)^2 + 2c(x - x_0)(x - x_1),$$

代入 $(h_0^1)'(x_0) = 1$ 解得

$$c = \frac{1}{(x_0 - x_1)^2},$$

因此

$$h_0^1(x) = (x - x_0) \left(\frac{x - x_1}{x_0 - x_1}\right)^2,$$

同理可得

$$h_1^1(x) = (x - x_1) \left(\frac{x - x_0}{x_1 - x_0}\right)^2.$$

□

4. 定义

$$(f, g) := \int_a^b f(x)g(x) \, dx,$$

对分片线性插值基函数 $\{l_j(x)\}$, 计算

$$(\text{刚度矩阵}) \quad S = (S_{ij})_{0 \leq i, j \leq k}, \quad S_{ij} = (l'_i, l'_j),$$

和

$$(\text{质量矩阵}) \quad M = (M_{ij})_{0 \leq i, j \leq k}, \quad M_{ij} = (l_i, l_j).$$

解答.

直接计算可得

$$\begin{aligned}
 S_{ii} &= \int_{x_{i-1}}^{x_i} \left(\frac{1}{h_{i-1}} \right)^2 dx + \int_{x_i}^{x_{i+1}} \left(-\frac{1}{h_i} \right)^2 dx = \frac{1}{h_{i-1}} + \frac{1}{h_i}, \quad \forall 1 \leq i \leq k-1, \\
 S_{00} &= \frac{1}{h_0}, \quad S_{kk} = \frac{1}{h_{k-1}}, \\
 S_{i,i+1} &= S_{i+1,i} = \int_{x_i}^{x_{i+1}} -\frac{1}{h_i} \cdot \frac{1}{h_i} dx = -\frac{1}{h_i}, \quad \forall 0 \leq i \leq k-1, \\
 S_{ij} &= 0, \quad \forall j \neq i-1, i, i+1,
 \end{aligned}$$

并且

$$\begin{aligned}
 M_{ii} &= \int_{x_{i-1}}^{x_i} \left(\frac{x - x_{i-1}}{h_{i-1}} \right)^2 dx + \int_{x_i}^{x_{i+1}} \left(\frac{x_{i+1} - x}{h_i} \right)^2 dx = \frac{1}{3}(h_{i-1} + h_i), \quad \forall 1 \leq i \leq k-1, \\
 M_{00} &= \frac{1}{3}h_0, \quad M_{kk} = \frac{1}{3}h_{k-1}, \\
 M_{i,i+1} &= M_{i+1,i} = \int_{x_i}^{x_{i+1}} \frac{x_{i+1} - x}{h_i} \cdot \frac{x - x_i}{h_i} dx = \frac{1}{6}h_i, \quad \forall 0 \leq i \leq k-1, \\
 M_{ij} &= 0, \quad \forall j \neq i-1, i, i+1,
 \end{aligned}$$

由此分别构成 2 个三对角矩阵.

□